CO2_AT1_MCQ TEST

Score Report and Correct Answers

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Total Score: 20 / 20

1. In informed search strategies, which algorithm uses only the heuristic value to select the node that appears closest to the goal?

Correct answer: Greedy Best-First Search

2. What is the evaluation function used in A* search?

Correct answer: f(n) = g(n) + h(n)

3. What property of a heuristic ensures optimality in A* search?

Correct answer: Admissibility

4. Which of the following is a local search algorithm?

Correct answer: Hill Climbing

5. Which algorithm is commonly used for decision-making in adversarial games?

Correct answer: Minimax Algorithm

6. A delivery robot chooses paths based only on estimated distance to the destination. Which algorithm is it using?

Correct answer: Greedy Best-First Search

7. Which algorithm combines the distance already travelled with an estimate of the remaining distance?

Correct answer: A* Search

8. Exam scheduling with constraints such as no overlapping exams and limited rooms is best modeled as what?

Correct answer: Constraint Satisfaction Problem

9. An AI chess program anticipates the opponent's responses and evaluates game states. Which algorithm is applied?

Correct answer: Minimax Algorithm

10. Which agent is suitable for exploring an unknown environment and updating knowledge while moving?

Correct answer: Online Search Agent

11. Why is an admissible heuristic important in A* search?

Correct answer: It ensures optimality

12. What is the primary benefit of Alpha-Beta pruning?

Correct answer: Reduces number of nodes evaluated

13. What is the main drawback of Hill Climbing?

Correct answer: Getting stuck in local maxima

14. What ensures a valid Constraint Satisfaction Problem (CSP) solution?

Correct answer: Constraint satisfaction

15. What is the role of an evaluation function in adversarial games?

Correct answer: Estimate utility of a state

16. In A* search, if g(n) = 7 and h(n) = 5, what is f(n)?

Correct answer: 12

17. What may happen if a heuristic overestimates the actual cost to reach the goal?

Correct answer: A* becomes non-optimal

18. For a full binary search tree with branching factor 2 and depth 4, what is the total number of nodes?

Correct answer: 31

19. In a minimax tree with branching factor 3 and depth 3, how many leaf nodes are present?

Correct answer: 27

20. What is the best-case time complexity of Alpha-Beta pruning?

Correct answer: O(b^(d/2))